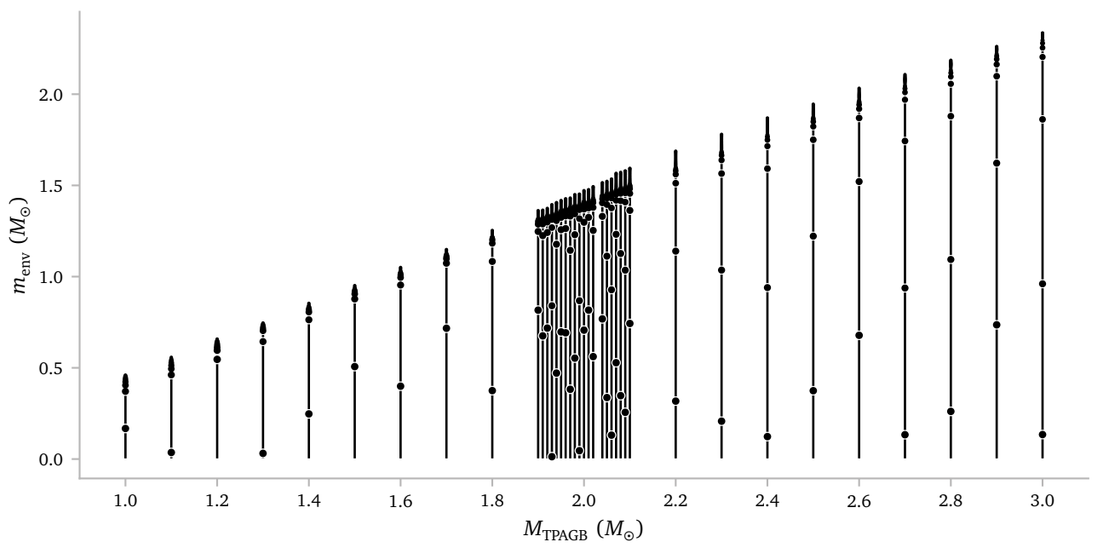
Koen Essers • Master Internship • November 8, 2026

Notes Week 36

a. Dense Mass Spacing Results

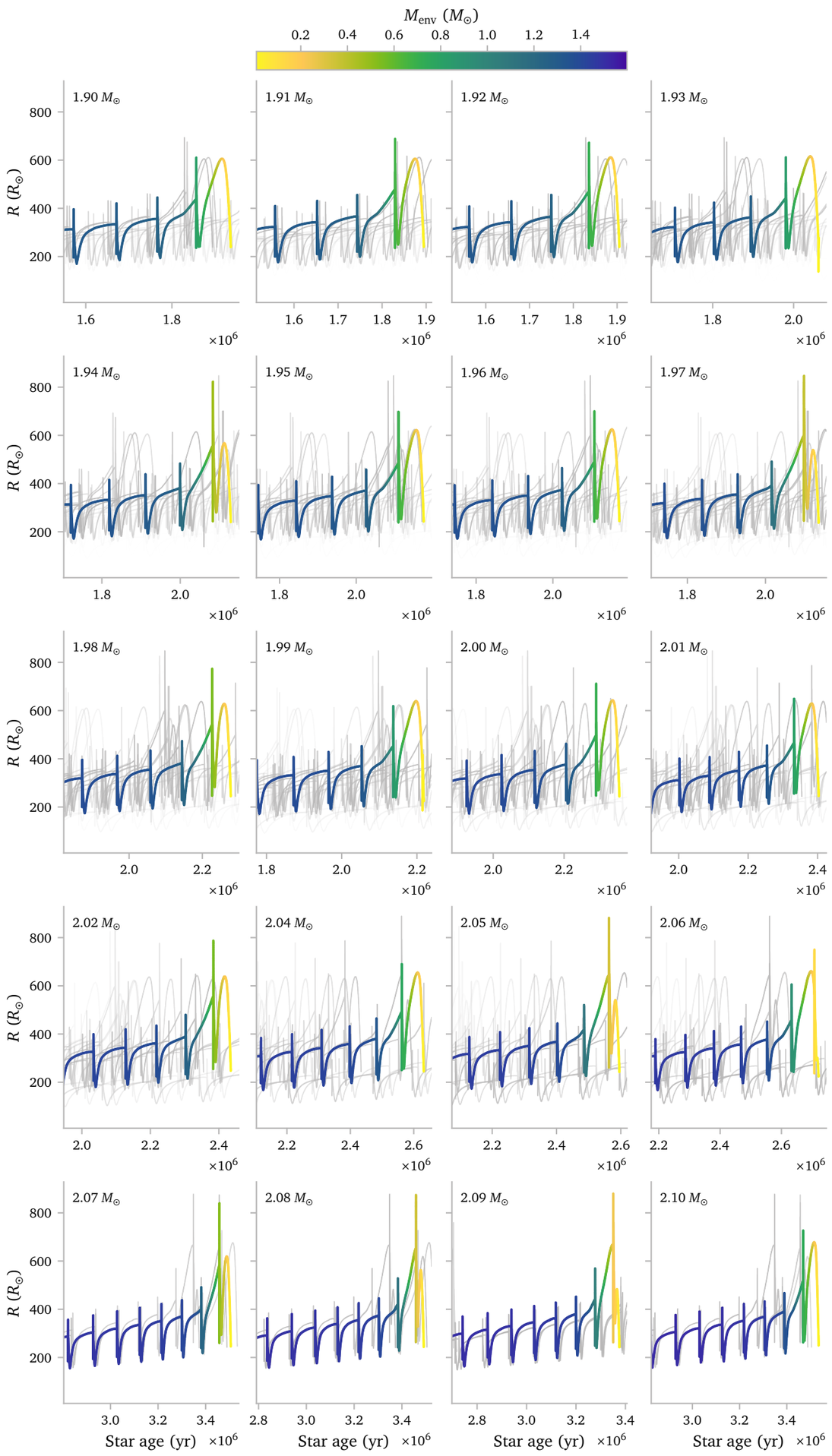
Below I plot the envelope mass for the new models and the old ones.



Quite clearly, even for 0.01 $M_\odot$ differences in mass, there is no predictable relation between changes in initial mass and envelope mass during the last thermal pulse.

It seems to be very “chaotic”. Small perturbations in initial mass lead to quite drastically different final TPAGB evolutions.

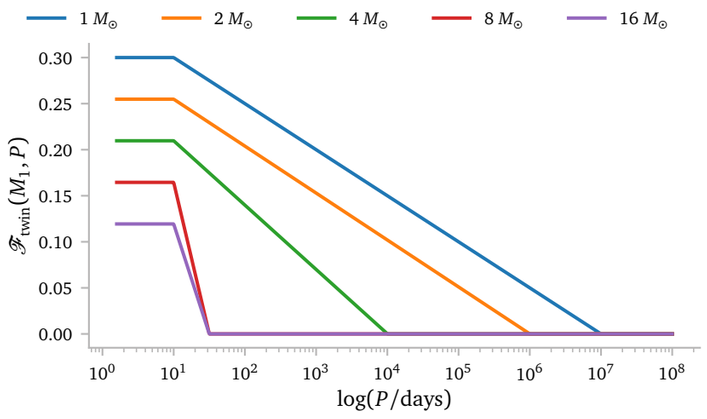
To highlight this, let me show the last few thermal pulses for the different masses between 1.9 $M_\odot$ and 2.1 $M_\odot$.



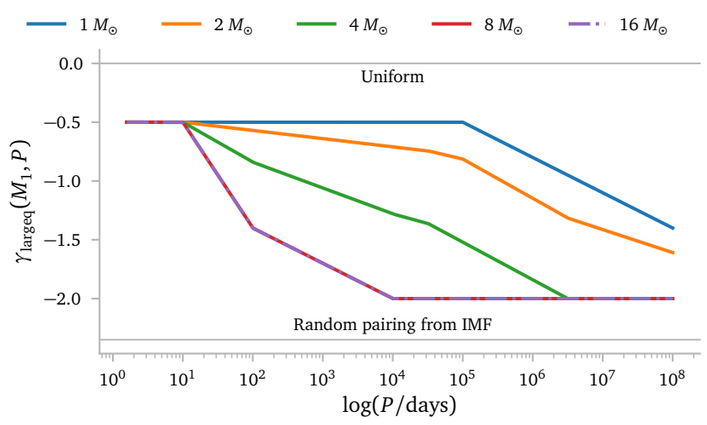
References

Moe, M. & Di Stefano, R. (2017) *Mind Your Ps and Qs: The Interrelation between Period (P) and Mass-ratio (Q) Distributions of Binary Stars*. ApJS230, 15.

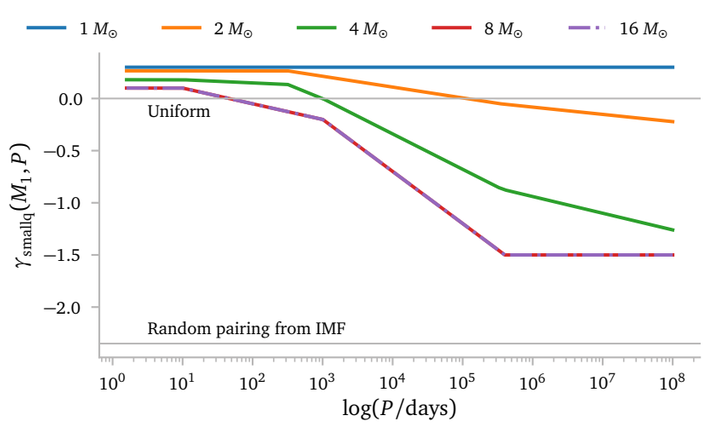
A Moe and Di Stefano Tests



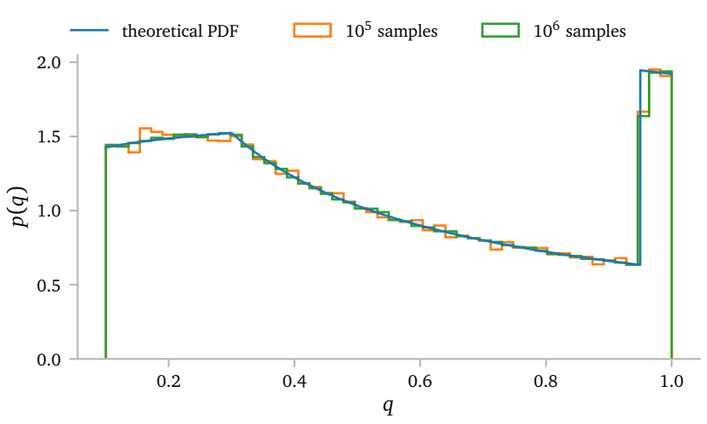
Compared with Figure 35 from Moe & Di Stefano (2017), these results look sensible.



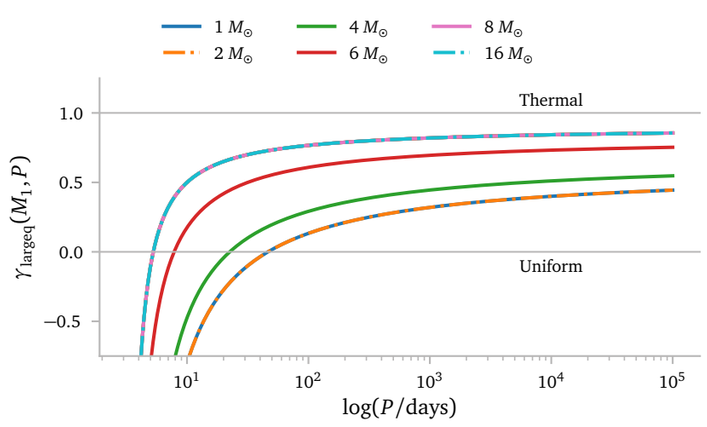
Compared with Figure 35 from Moe & Di Stefano (2017), these results look sensible.



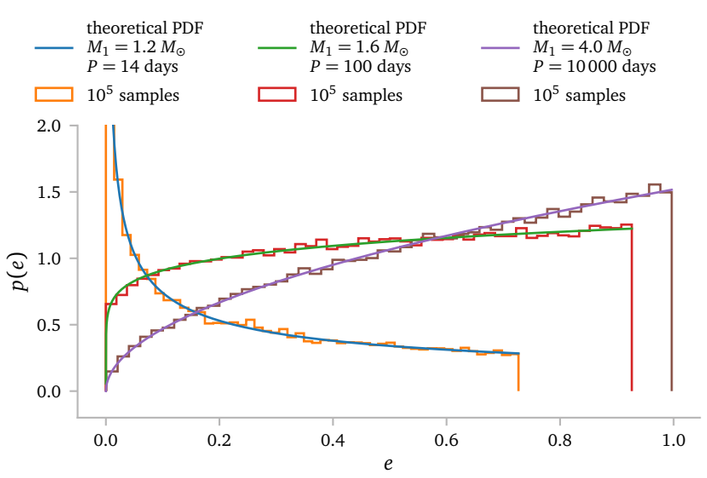
Compared with Figure 35 from Moe & Di Stefano (2017), these results look sensible.



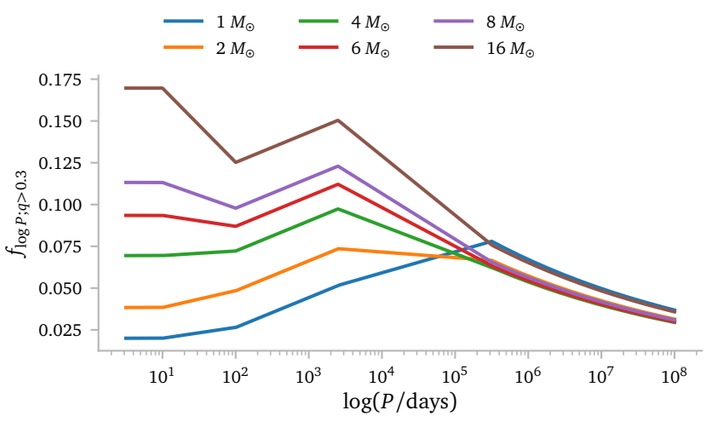
Compared with Figure 2 from Moe & Di Stefano (2017), these results look sensible.



Compared with Figure 36 from Moe & Di Stefano (2017), these results look sensible.



Based on how power laws should look and the figure above, I think this makes sense.



Compared with Figure 37 from Moe & Di Stefano (2017), these results look sensible.